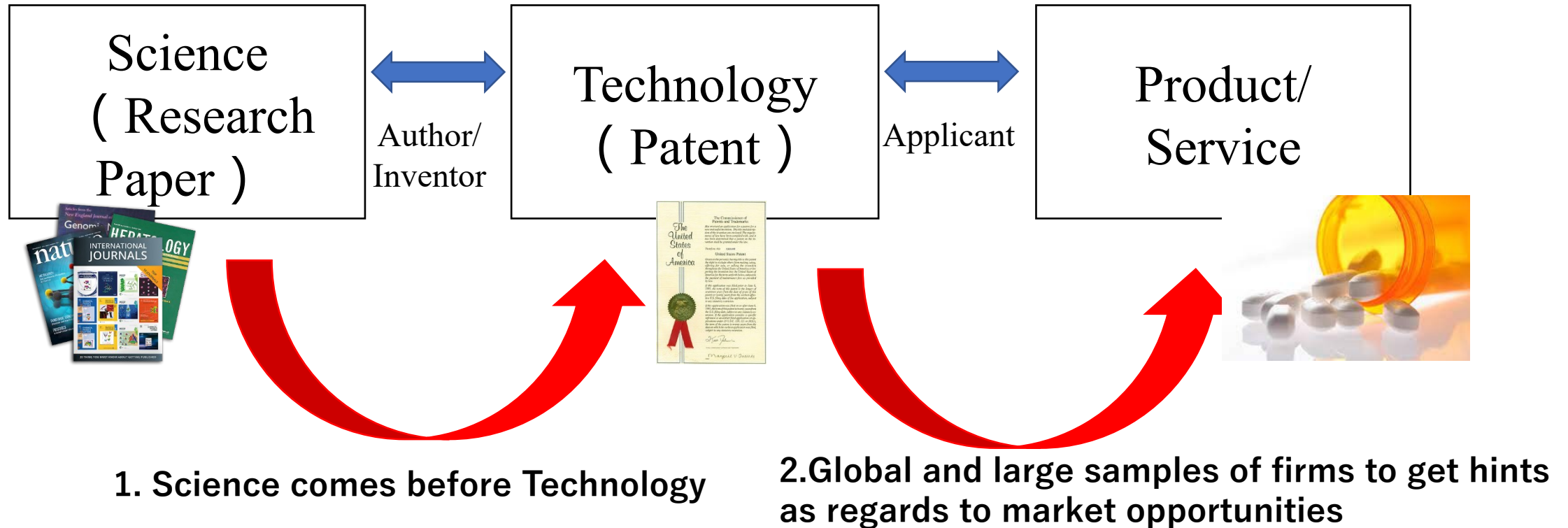


Introduction to Phase 2

What is innovation opportunity discovery?

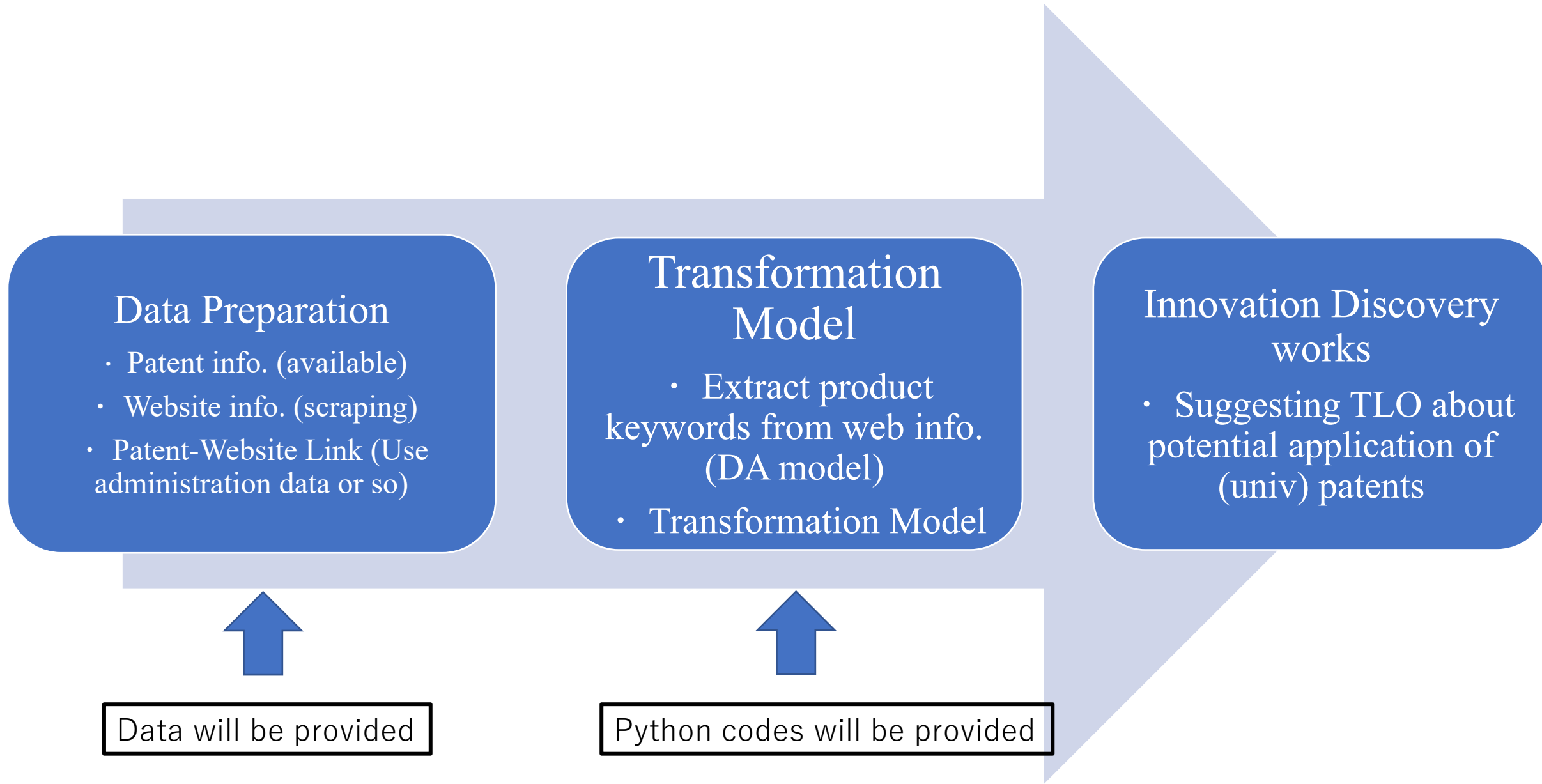


Machine learning model

to predict potential technology/market (innovation) opportunities

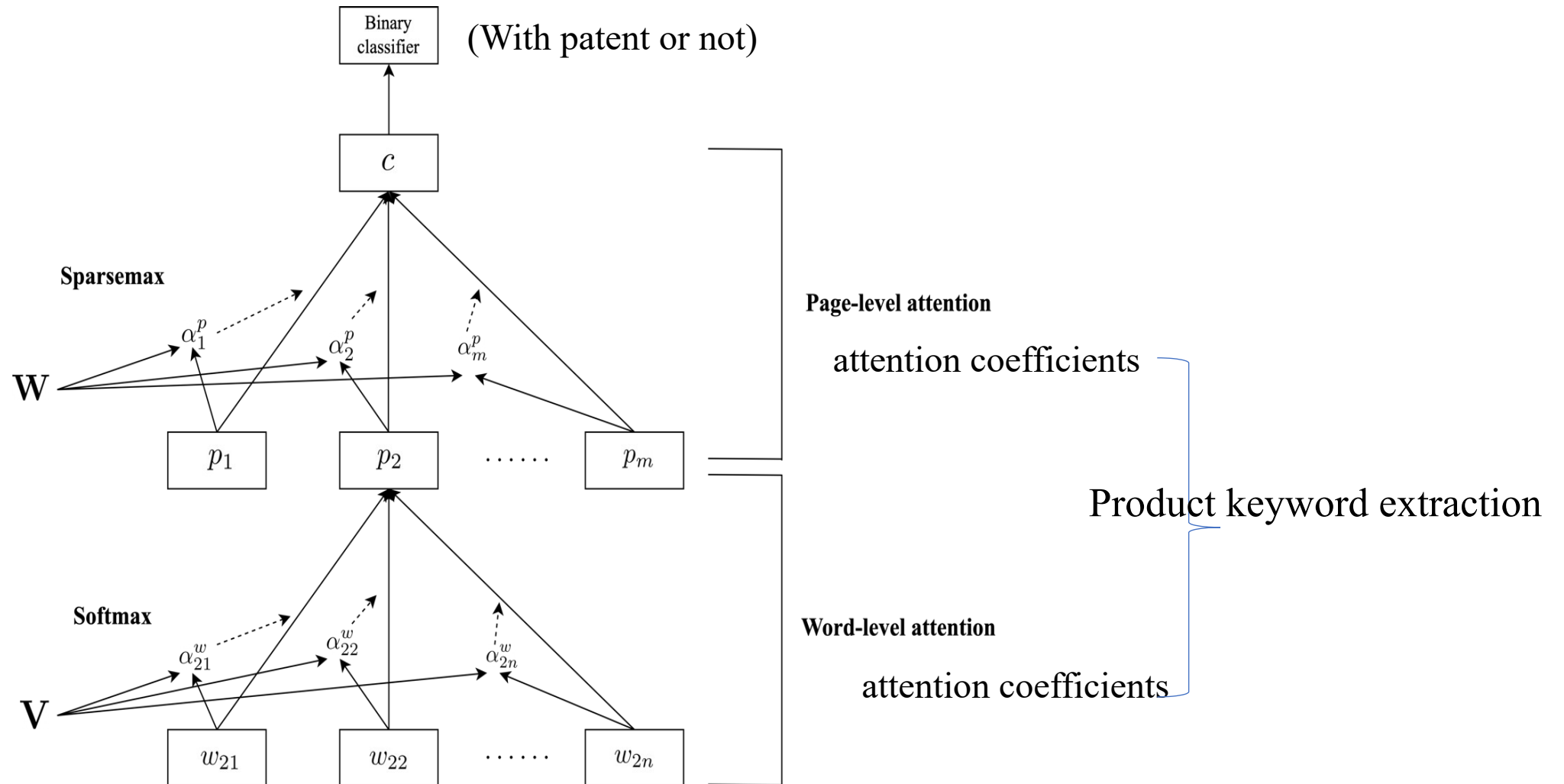
Youtube video for detail : <https://www.youtube.com/watch?v=Y6TIHxfKsmM>

Process of works



Dual attention model for keywords extraction

(Skimming product information from company web information)



Motohashi, K. and Zhu, C., Identifying Technology Opportunity Using a Dual-attention Model and a Technology-market Concordance Matrix, forthcoming *Technology Forecasting and Social Change*

Example of Dual Attention Model

Hakuto Co. Ltd (Electronics Device Distributor)

Page level attention

The Web portfolio of a randomly selected company	Attn-weights
http://www.hakuto.co.jp/irinfo/	0.0
http://www.hakuto.co.jp/irinfo/announce/	0.0
http://www.g5-hakuto.jp/	0.0
http://www.hakuto.co.jp/profile/ethic/	0.0
http://www.hakuto.co.jp/news/2022/20221024.html	0.0
http://www.hakuto.co.jp/news/2022/20221111.html	0.0
http://www.hakuto.co.jp/products/equipment/	0.41
http://www.hakuto.co.jp/profile/outline/strategic/index.html	0.0
http://www.hakuto.co.jp/profile/	0.0
http://www.hakuto.co.jp/products/components/	0.1
http://www.hakuto.co.jp/profile/outline/embedded/index.html	0.0
http://www.hakuto.co.jp/news/2022/20221102.html	0.0
http://www.hakuto.co.jp/news/2022/20221013.html	0.0
http://www.hakuto.co.jp/eco/	0.0
http://www.hakuto.co.jp/products/	0.0
http://www.hakuto.co.jp/site_map/	0.0
http://www.hakuto.co.jp/news/2022/20221020.html	0.0
http://www.hakuto.co.jp/contact/	0.03
http://www.hakuto.co.jp/profile/outline/advanced/index.html	0.0
https://www.process.hakuto.co.jp/	0.18
http://www.hakuto.co.jp/	0.0
http://www.hakuto.co.jp/privacy/	0.0
http://www.hakuto.co.jp/products/chemicals/	0.27
http://www.hakuto.co.jp/products/devices/	0.0
http://www.hakuto.co.jp/sitepolicy/	0.0
http://www.hakuto.co.jp/news/	0.0
http://www.hakuto-vacuum.jp/	0.0
http://www.hakuto.co.jp/news/	0.0
http://www.hakuto.co.jp/	0.0
http://www.hakuto.co.jp/news/2022/20221115.html	0.0
http://www.hakuto.co.jp/irinfo/	0.0

Product page



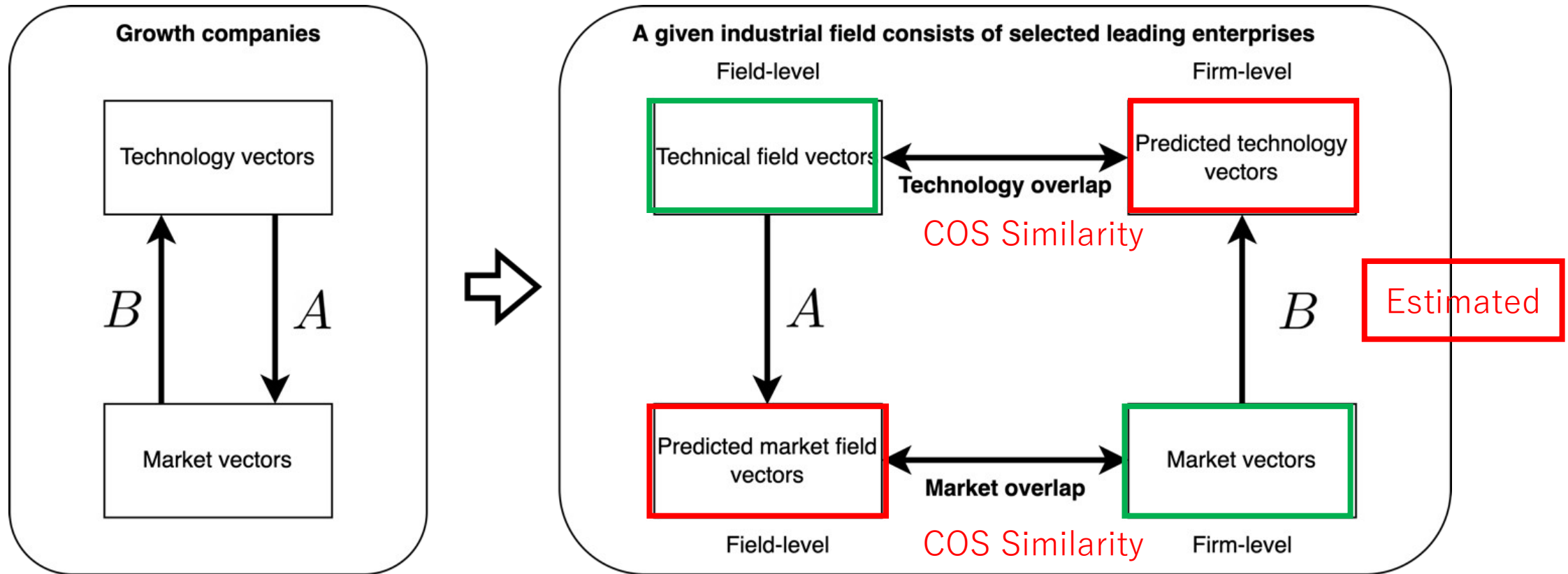
word level attention

Extracted keywords (translated words)

太陽 (solar) 機器 (machine) デバイス (device) テクノロジー (technology) イオン (Ion) 電気 (electronic) スタンダード (standard) 薄膜 (film) コンポーネント (component) 光通信 (Optical communication) ケミカル (chemical) 材料 (material) 製品 (product) 装置 (device) 成形 (mold) 探す (search) 試験 (experiment) メーカー (manufacturer) 電気 (electronic) ケミカル (chemical) 粘着 (adhesion) 製品 (product) ブランド (brand) 機器 (machine) デバイス (device) ボイラ (boiler) 精製 (purification) フェルト (felt conditioner) 薬品 (drugs) ODM コンポーネント (component) 研究 (research) PVD 機器 (machine) 問合せ (inquiry) Compact CL モジュール (module) 起こす (induce) MS situ 生産 (production) エッチング (etching) ガス (gas) FA 量産 (mass production) 制御 (control) 材料 (material) イオンビームミリング (Ion beam milling) 製品 (product) シリーズ (series) 装置 (device) 半導体 (semiconductor) セル定性 (qualitative) 用途 (usage) 完結 (completion) 有機 (organic) MBE CVD リフロー (reflow) LTI PVA ブレイクスルー (breakthrough)

Transformation model and its usage

Use of new firm's alignment information for opportunity identification



Introduction for Phase 2

(Goal) Suggesting product/service for patents by using technology-product converting model

(Dataset)

- Firm's web text + patent info for three countries (CN, JP, US) (will be supplied by Tong Xinyu)

(Process)

- Topic modeling of tech (patent info.) to see the technology evolution
- Extract product related keywords by dual attention model from scraped website information
- Develop technology-product converting model and come up with suggested products based on some existing patents

To do by timeline

- 1st Week : June 16-20 :
 - Topic modeling, understanding technical contents
 - Start scraping firm's webpage in 2018 (wayback machine) -> Meeting June 23, 12:00- JST
 - 2nd-3rd Weeks : June 23-July 4 :
 - Dual attention model + transformation matrix face validation -> Meeting **July 3**, 12:00-JST
 - 4th-5th Weeks : July 7-July 18 :
 - Time difference model : topic modeling of 2013-18 patents, 2018-23 patents, 2018 HP and 2023 HP, any change?, Model building 2013-18 P -> 2018 HP and 2018-23 P -> 2023HP) -> Meeting July 14 and July 21, 12:00-JST
 - 6th-7th Weeks : July 21-Aug 1 :
 - Methodological development -> dual attention model (methodological development, use of existing corpus such as Wikipedia), transformation model (not linear projection)
 - Demo-day Aug 1 12:00- JST
- * **Please note that July 3 is Thursday, not Monday!**